PERFORMANCE WORK STATEMENT

Power System Modeling and Analysis
23 March 2026

1 SCOPE

The Contractor shall provide comprehensive, non-personal Power System Modeling and Analysis services to ensure the reliability and safety of electrical distribution systems at the locations specified below. The Contractor shall not perform any inherently governmental functions, as defined in FAR 7.503. The Contractor shall furnish all labor, software, materials, tools, equipment, transportation, and supervision necessary to:

- Collect field data related to the electrical distribution system.
- Develop and/or update a comprehensive electrical model of each installation's power system.

Perform the following power system studies using the electrical model:

- Power Flow Analysis
- Cross Connect Capability Evaluation
- Short Circuit Study
- Single Point of Failure (SPOF) for All Generator Locations
- Protective Device Coordination Study
- Arc Flash Hazard Analysis
- Distribution System Reliability

Generate detailed analysis reports documenting the findings and recommendations for each study, including mitigation strategies for identified deficiencies. The executive summary shall include the deficiencies with the remedies. The executive summary shall also include a simplified one-line with cross-connect capabilities outlined and at each switch.

The services shall be performed at the following installations:

Group A:	Group B:	Group C:
MacDill AFB (Florida)	Nellis AFB (Nevada)	Moody (Georgia)
Patrick SFB (Florida)	Fairchild AFB (Washington)	Clear SFS (Alaska)
Cape Canaveral SFS(Florida)	Holloman AFB (New Mexico)	Aviano AB (Italy)
Peterson SFB (Colorado)		Gimhae AB(South Korea)
Incirlik AB (Turkey)		Suwon AB (South Korea)
Misawa AB (Japan)		Arnold AFB (Tennessee)
Yokota AB (Japan)		Moron AB (Spain)
Cannon AFB (New Mexico)		

The electrical distribution systems to be analyzed include, but are not limited to switching stations, substations, transmission/distribution/ service cables, primary distribution equipment, government-owned power generating stations, government-owned renewable power generation (e.g., wind, solar), and generators with automatic or manual transfer switches.

The Contractor shall conduct a comprehensive analysis of the base's electrical system, from the primary power distribution lines through all secondary circuits, terminating at the service entrance disconnect of each building. This analysis will include field assessments of all assets within the defined scope, except as explicitly excluded below.

Elements Included in Analysis:

The analysis shall specifically address the following:

- **Medium-Voltage and Primary Distribution Equipment:** Assess all medium-voltage and primary distribution equipment located though out the base and within the facilities.
- **Secondary Low-Voltage Distribution Equipment**: Include assessing all low-voltage and secondary distribution equipment located throughout the base and within the facilities to the main panel disconnect.
- **Emergency and Backup Generators:** Integrate emergency and backup generators into the system model. The Air Force Civil Engineer Center (AFCEC) will provide a database of these generators. The Contractor shall validate the accuracy of the database information through on-site inspections.
- **Electric Motors:** Analyze all electric motors rated at 50 horsepower (HP) or greater connected to any point within the defined distribution system.

Exclusions:

- **Military Family Housing:** The analysis excludes all portions of the distribution system owned and operated by Military Family Housing, unless otherwise specified in Attachment 1 - Base Specific Requirements.

The Contractor shall utilize existing government-provided teleconferencing platforms for all scheduled meetings to facilitate participation by off-site personnel. If existing platforms are unavailable, the Contractor shall provide a compatible alternative. The Contractor shall schedule meeting with the AFCEC SME Electrical Engineer every 2 months to discuss the project progress and status.

2 EXISTING DISTRIBUTION SYSTEMS

All information provided for each installation's existing distribution system is based on previous distribution system studies, if available, and interviews with installation personnel. The base specific information included in Attachment 1 - Base Specific Requirements may not be fully up to date. Refer to the attached documents for additional information.

3 PERFORMANCE REQUIREMENTS

All work shall be coordinated with Base Civil Engineering (BCE) personnel - Base Electrical, Base Planner, Base Programmer, Base Real Property, Base Power Pro, Base Electrical Shop, Base Installation POC and Base Project Manager. The Contractor shall coordinate all work with the BCE a minimum of 30 calendar days CONUS and 90 calendar days OCONUS in advance of any site visits. The scope for this study shall include the following work items to be performed for each installation.

Study consists of analyzing the existing electrical distribution system and determining proper modifications and extensions of the overhead and underground electrical distribution systems to improve safety, efficiency, maintainability, reliability, and to support present and projected electrical requirements. The project shall include the following general activities:

- Preliminary Data Gathering........*(refer to Section 3.1)*
- Field Data Gathering..................*(refer to Section 3.2)*
- Build/Update Model....................*(refer to Section 3.3)*
- System Analysis.........................*(refer to Section 3.4)*
- Draft Report...............................*(refer to Section 3.5)*
- Follow-up Data Gathering...........*(refer to Section 3.6)*

- Final Report.................................*(refer to Section 3.7)*
- Corrected Final Report...............*(refer to Section 3.8)*
- Equipment Labeling.....................*(refer to Section 3.9)*
- Software and Training.................*(refer to Section 3.10)*

3.1 PRELIMINARY DATA GATHERING

The Contractor shall provide field verification and data collection of all information necessary to perform all components of the System Analysis, such as the following information.

- All equipment and protective devices located within each substation
- All equipment and protective devices located within each switching station
- All equipment and protective devices located within each generating station
- Nameplate data for all transformers
- Nameplate data for all switches
- Nameplate data for all sectionalizing cubicles
- Nameplate data for all facility service disconnects
- Nameplate data for all power generation equipment (both renewable and non-renewable) feeding directly into the primary distribution system and buildings.
- Nameplate data for all power generation equipment attached to the facility by either ATS or MTS
- Nameplate data for all ATS or MTS connecting power generation equipment to a facility
- Cables sizes and types
- All motor loads, including motor starting type, on the installation over 50HP
- Field verification of all overcurrent protective device (OCPD) information and settings
- Verify that the facility's main electrical distribution panel has a surge protection device
- Verify that the facility's electrical distribution panel room has a common ground
- Verify Lightning arresters are installed and if found place on one-line
- Verify Fault indicators and place on one-line if found on equipment.
- Coordinate any base dewatering efforts of manholes.

The Contractor shall be responsible for coordination of all permits and activities necessary to collect the required information such as outages, energized work permits, confined space entry, secure area entry, etc. The Contractor shall also be responsible for providing all equipment and tools needed to facilitate access to equipment and recording of information.

Upon contract award, the Contractor shall review the existing distribution map, existing one-line diagram, existing model, and the most recent study report (if they exist) and develop a list of information and documents that will be required to update the model and perform the system analysis. The information contained within the existing distribution map, existing one-line diagram, existing model, and the most recent study report (if they exist) shall not be assumed accurate and shall require field verification to ensure that erroneous information is not used in the analyses, refer to Section 3.2. Once the list of required information and documents has been created, the Contractor shall initiate contact with the BCE.

Through coordination with the BCE, the Contractor shall attempt to collect as much of the needed information and documents as possible prior to any site visits. The Contractor shall attempt to collect the most up-to-date versions of all available electrical system drawings, lists of projects under construction and planned, and power meter readings and historic load information prior to the start of field data gathering activities. The Contractor shall also collect all available information from the BCE regarding all distribution equipment modifications which have occurred since completion of the previous analysis (if a prior analysis has been completed).

The Contractor shall request copies of proposed/planned/existing projects, real property records list of facilities. All facilities list in the real property record must be included in the base evaluation, model, drawings, etc. Existing facilities upgrades and new facilities projects must be included in the base evaluation, model, drawings, and GIS data. The proposed/planned projects must be included in the base electrical model (5 Year Plan). For additional assistance contact AFCEC SME Electrical Engineer.

The Contractor shall request a detailed description the normal operating configuration of the existing distribution system. As part of this description the Contractor shall verify if the normal operating configuration is routinely varied do to normal base operations, seasonal changes, routine maintenance, or any other factor other than emergency maintenance.

The Contractor is responsible for gathering crucial information regarding the base's electrical infrastructure from the Base Civil Engineer (BCE) office.

1. **Electrical System Inventory and Critical Infrastructure:** Mission Owners with Base Civil Engineer support will identify their Critical Facilities that host Mission Essential Functions. They will list the function, associated facility number, and an unclass description of the function. Then they will state what the Maximum Tolerable Downtime (MTD) in minutes is for that essential function. BCE, with AFCEC/COSM analysis support, will compare the required MTD with actual electricity availability for that facility. Thenrecommend projects or plans that can be determined to mitigate any shortfalls in power availability.

Mission Essential Function ID	Building Number	Maximum Tolerable Downtime (In Minutes)	Mission Description (Unclassified)	OSD Availability or Reliability Standard
MEF-06	10344	32	Data Processing	99.9%

2. **Planned Electrical Load Modifications:** The Contractor shall solicit information regarding all planned construction projects or anticipated changes to the electrical load on the system scheduled to occur within the next five years. This data is typically documented in the base's 5-Year Plan. Prior to concluding on-site data collection (as detailed in Section 3.2), the Contractor is obligated to obtain an updated copy of this 5-Year Plan from the BCE. This requirement ensures the study's accuracy by incorporating any foreseeable modifications to the electrical system.

Contact with any utility companies supplying the installation shall also be initiated during the preliminary data gathering phase. At a minimum the Contractor shall obtain the utility's available fault current information for each point of connection to the base owned distribution system, power outage history, and any unusual system design information (including nearby power factor correcting capacitors) and a one-line drawing of their substation(s). The Contractor shall also request all historic load information available from the utility companies. All enquiries to and from the utilities shall be made in English.

The Contractor shall coordinate with the BCE for a complete and updated list of any security/sensitive locations within the installation which would require special permission, special escorts, and/or restrictions to access. It may not be possible to resolve all items required to gain access to security/sensitive location within the installation during preliminary data gathering; if necessary, coordination of access to these locations shall resume during the first field data gathering site visit. The time needed to gain authorization to access security/sensitive locations varies by location and can take an appreciable amount of time in some instances. The Contractor shall exhaust all legal options working with the personnel at these security/sensitive locations. In some locations the Contractor will need additional time to gain authorized access per Attachment 1 - Base Specific Requirements. If access is still not permitted, the Contractor shall contact the Contracting Officer's Representative (COR).

3.2 FIELD DATA GATHERING

Any information or documents in possession of the BCE which could not be obtained during preliminary data gathering shall be collected during the Contractor's site visits. All information and documents received from the BCE shall be field verified by the Contractor to ensure that erroneous information is not used in the analyses. All errors discovered in the information and documentation provided by the BCE shall be reported to the BCE within one calendar day.

During the first day of the first field data gathering site visit, the Contractor shall coordinate and attend an on-site Coordination/Kick-Off meeting with the Base Project Manager and the Government points of contact (POCs) provided by AFCEC. The purpose of the meeting includes items such as scope/contract discussions, project tasks, project status, and the general exchange of information concerning past projects, current and future activities. The Base Project Manager shall furnish available information and discuss operating conditions and base personnel's specific concerns. The Contractor shall provide a meeting agenda to everyone indicated below to receive meeting minutes. The Contractor shall facilitate the meeting, record attendance, and record meeting minutes. The Contractor shall distribute meeting minutes to each installation, COMMAND, AFCEC, and the USACE POC by close of business the following business day. (CDRL A005)

Field data gathering shall include determination and verification of all existing conditions, including primary feeder runs, locations, manufacturer and types of equipment in the electrical system, transformer and large motor nameplate data, protective device characteristics, total connected KVA Transformers on each feeder, and other pertinent information necessary to verify and update existing documents and models to reflect actual conditions. The Contractor shall be responsible for collecting all information necessary for the completion of all components of the System Analysis, refer to Section 3.4. Furthermore, the Contractor, once gaining access inside equipment, shall complete all assessment and data collection before closing and proceeding to the next piece of equipment. Equipment information gathered during the site visit shall include items such as:

- All Equipment Manufacturer and Nameplate Information (manufactory name, manufactory date, model number, serial number) – include GIS mode
- Equipment IDs – include in GIS Model
- Type of Equipment – include GIS model
- Size of Equipment – include GIS model
- Equipment Voltage – include GIS model
- Current Transformer (CT) Ratios (A/FLA/FLC) – include GIS model
- Phase Signal for 3 Phase – include GIS model
- Power Rating (KW, KVA, HP (for motors) – include GIS model
- Standard Certification (UL, IEEE, ANSI, NEMA, IEC)) – include GIS model
- Transformer Tap Settings – include GIS model
- Radial feed, feed through (how is connected primary and alternate paths)
- Equipment grounding and type of grounding – include GIS model
- Lightning Protection Type – include in GIS Model
- Reclosure Settings – include in GIS Model
- Relay/Breaker Settings – include in GIS Model
- Breaker Sizes – include in GIS Model
- Fuse Sizes – include in GIS Model
- Feeder Conductor Sizes – include in GIS Model
- Service Conductor Sizes – include in GIS Model
- Conductor Lengths – include in GIS Model

- Pole Numbers and Condition of at least 30 poles (randomly selected) – include in GIS Model
- Manhole/Handhole IDs – include in GIS Model
- Emergency Generator and size – include in GIS Model
- Global Positioning Systems (GPS) Coordinates
- General Condition of Equipment *(Example: Green+, Green, Green-, Amber+, Amber, Amber-, Red+, Red, Red- on a 100-0 scale, Green+ is considered brand new installation while red- is required replacement within a year or inoperable)*
- Measurement of enclosure for Arc Flash Calculations
- Short Circuit Current Ratings
- ATS and MTS ratings
- Generator control circuits and governor settings.
- Facility square footage and number of facility occupants.

Mapping Grade GPS or comparable traditional survey methods will be used to collect geospatial data (e.g., northing and easting) for all contract activities where geospatial data is involved.

Existing power meters can be used only when all information above can be directly readable and base electrical personnel can attest to the accuracy of the existing power meters

Base electrical personnel from each installation will be provided by the Government when required for escort during the fieldwork.

All Contractor personnel shall use normal base working hours of 0800 to 1700 Monday to Friday (or as indicated in Attachment 1 – Base Specific Requirements) and exclude any federal holidays unless coordinated in advance or waved by the BCE, or BCE representative. Base keys are not assumed to be checked out to the Contractor unless waved by Attachment 1 – Base Specific Requirements, BCE, or BCE representative.

To allow the BCE adequate time to ensure access to each facility, the Contractor shall provide the BCE with a list of facilities requiring entry a minimum of 20 working days prior to each site visit. Advance notice more than 20 working days may be required to ensure access to some facilities, refer to Attachment 1 – Base Specific Requirements. The Contractor shall receive confirmation from the BCE prior to the site visit that access will be granted to each facility. If, for any reason, access cannot be granted to a facility during a site visit the Contractor shall coordinate with the BCE to schedule access to the facility at a later date.

As part of the monthly status report submitted prior to the final scheduled field data gathering site visit, the Contractor shall present general findings of the field surveys and discuss the impressions resulting from the initial phases of the engineering evaluations and system analyses. The Contractor shall discuss anticipated problems if portions of the tasks appear questionable or unsound because of the data gathering and initial evaluation efforts.

3.2.1 Inaccessible Equipment

If, during of the field data gathering site visit, access to a piece of equipment is found to be impeded, the Contractor shall immediately notify the BCE. Working together with the BCE, the Contractor shall make every attempt to gain access to the equipment, such as contacting appropriate individuals with keys, cutting locks (with the approval of the BCE), using bucket trucks, performing outages, applying for hot work permits, and entering confined spaces.

The Contractor shall be required to provide all tools and equipment necessary to facilitate access to distribution equipment. These items shall include power tools, hand tools, appropriate personal protective equipment (PPE), and bolt cutters. At least one (1) of these employees shall be trained and licensed to record the information indicated above. The Contractor shall be responsible for coordinating any necessary outages or obtaining any energized work permits required. The Contractor shall exhaust all legal options working with the personnel at security/sensitive locations to gain authorized access. If access is still not permitted the Contractor shall contact the COR.

A piece of equipment shall not be considered inaccessible due to being located within a locked room, within a padlocked enclosure, within a security/sensitive location, or in any other standard installation location or configuration. The Contractor shall have a telephoto camera, high powered binoculars, and flashlights for field inspection of equipment located on poles and within manholes. If the Contractor is unable to access any piece of equipment, despite having taken all reasonable actions to gain access, approval shall be required from the COR to abandon attempts to access the equipment. The COR shall be notified immediately via email and telephone, if the Contractor is unable to access any piece of equipment. Upon notifying the COR of impeded access to a piece of equipment, the Contractor shall proceed with field data gathering activities for the remaining equipment until a response has been received from the COR. If, after two (2) working days, the COR has not provided the Contractor with a decision the Contractor may declare the equipment inaccessible. The Contractor shall document all attempts to obtain clarification from the COR regarding the accessibility of equipment. Any equipment which has been declared inaccessible shall be entered into the model using appropriate and clearly documented assumptions.

3.2.2 Acceptable Assumptions

Assumptions based on accepted industry practice and published standards shall be acceptable if they are clearly documented. (*example: assuming transformer x/r ratio based on IEEE standards*). All assumptions for conductors or protective devices that are based on industry standard practice or published standards shall have documented BCE and electrical shop acceptance.

If it is not possible to field verify the information for bayonet fuses within an oil-filled transformer without physically removing the fuses, it shall be acceptable for the Contractor to use assumed values or values provided by the BCE for the fuse information. The Contractor shall coordinate with, and receive concurrence from, the BCE and the Base Electrical Shop when determining acceptable assumptions to use for bayonet fuse values. The Contractor shall document the full name, job title, email, and phone number of all individuals coordinated with to determine the acceptable assumptions for bayonet fuse values. This documentation shall be included in the monthly status report, refer to Section 3.2.7, immediately following the determination of the coordinated assumption. The documentation shall also be included in the CD's or DVDs submitted concurrently with the reports, refer to Section 3.2.8.

The Contractor shall utilize all information available and exercise sound engineering judgment when making any assumptions. Assumptions must be kept to a minimum and shall only be used where a device is deemed inaccessible, refer to Section 3.2.3, or where field verification of the information would require an outage or energized work which is specifically disallowed by the BCE.

Justification for all assumptions shall be included within the report, refer to Section 3.5.1 for additional information. All assumptions made by the Contractor are subject to approval or rejection by the government during the report reviews. Any assumption rejected by the government shall be field verified at no additional cost to the government, or a new assumption made in coordination with the COR and BCE.

3.2.3 Notification Requirements

The Contractor is required to notify the Contracting Officer (CO), COR, and BCE of critical issues that may affect the contract performance and/or human health and the environment. The types of issues that require notification include health risks, spills, unexpected utility crossings, unusual weather conditions, unacceptable materials, and changes in critical personnel. On critical issues, verbal notification should be made immediately, followed by written notification as soon as practical.

3.2.4 Work Site Coordination

The Contractor shall coordinate with the COR and BCE to arrange access to all facilities. The Contractor shall comply with Occupational Safety and Health Administration (OSHA) safety and health regulations and local safety office requirements. The Contractor is required to provide the CO copies of any OSHA report(s) submitted during the duration of the contract.

3.2.5 Monthly Status Report (CDRL A003)

The Contractor shall provide a monthly status report which encompasses the efforts being performed at each installation which discusses the progress, a description of activities conducted since the previous monthly status report, schedule, and any issues. Any changes in the schedule shall be highlighted, and an accompanying justification shall be provided outlining the reasons for the change.

3.2.6 Dissemination of Collected Data (CDRL A007)

In addition to the information required to be included in the reports, all data collected shall be provided to the government on a separate CD(s) or DVD(s) as part of the Corrected Final Report, refer to Section 3.8. This shall include all documents, pictures, measurements, equipment data, GPS location data, etc. Data shall be stored in directories in an organized manner which allows subsequent users to readily locate and retrieve the information for any facility or equipment. Collected data will become the property of the government.

The Contractor shall provide digital photographs of all field conditions influencing conclusions in the submittal for each installation. Photographs shall be provided of all equipment validated, unless prohibited due to base security requirements. Where providing a photograph of a piece of equipment is not possible, a brief narrative shall be submitted describing the conditions of the equipment. One (1) digital photograph at each location shall be identified with a geographic coordinate and stored with the equipment nameplate data. All photographs shall be submitted with the Corrected Final Report, refer to Section 3.8. Digital photographs shall be of such quality and size to clearly show field conditions, equipment ratings, specifications, nameplate data and verify quantity and configuration of the associated equipment. Provide all photos using JPG format. Equipment nameplate photographs shall be linked to the associated device within the software model. Any equipment found to be in poor physical condition or having significant physical deficiencies shall have photographs showing the deficiency linked to the associated device.

3.2.6.1 Spatial Data Requirements

All maps and associated data must comply with the latest version of Spatial Data Standards for Facilities, Infrastructure, and Environment (SDSFIE) available from the SDSFIE web site. The SDSFIE web site address can be found at https://www.sdsfie.mil/. This data will be organized using the latest version of the SDSFIE standard, Air Force Adaptation, approved by the Installation Geospatial Information & Services (IGI&S) Governance Group (IGG) available from the Air Force Civil Engineer Center (AFCEC) Geo Integration Office (GIO) as the USAF lead for IGI&S capabilities. The SDSFIE standard is the authority for file and feature class identification and definition, attribution and valid domain values. When any geospatial information collected as a result of the contract includes information identified in the Common Installation Picture (CIP) or recognized Mission Data Set (MDS) the Contractor will deliver data consistent with the established requirements for the data and will ensure functionality with the receiving system. Information must be collected at no less than 1:1200 scale for base cantonment areas and 1:4800 scale for larger undeveloped base areas. Spatial data will meet or exceed National Map Accuracy Standards at those scales. Metadata will be provided and will use Federal Geographic Data Committee (FGDC) Content Standards for Digital Geospatial Metadata (CSDGM) for organization.

Geospatial data must be delivered in a geo-referenced Geographic Information (GIS) format (feature-based file structures with one-to-one cardinality between spatial records and attribute records) using Environmental Systems Research Institute's (ESRI) geodatabase format. All attribute data as specifically outlined in the task order contract must be included either in the GIS data file or as a separate table with a SDSFIE key variable that may be used to relationally join the separate table with the GIS data file. All geospatial data must be delivered using the projection, coordinate system, and coordinate units stipulated by the AFCEC GIO (e.g, WGS-84, North American Datum 1983 (NAD83) projection, State Plane Coordinate System, feet or metric coordinate units). Further guidance on mapping units, coordinate systems and projections are available from the AFCEC GIO and/or Installation GIO.

Mapping- or Survey-Grade GPS or comparable traditional survey methods will be used to collect geospatial data. The use of mapping- or survey-grade GPS will depend on the precision requirements of the product data. These requirements will be specified later in this Performance Work Statement (PWS) for all contract activities where geospatial data are involved. In the case of contracts involving utility construction, location and attribute data will be obtained at the time of excavation. Further information about precision requirements should be obtained from the AFCEC GIO or installation GIO.

One-Line diagrams included in this drawing package must adhere to the standards specified in this contract. Diagrams shall reflect the approximate geographical arrangement of equipment and be designed for seamless assembly into a consolidated 72” x 96” layout by printing and taping individual sheets. To ensure clarity and consistency, all labels, linework, and symbols must conform to standardized conventions, maintaining uniformity in font, size, and spacing. All symbols and text shall be clearly legible without magnification.

Diagrams should optimize the use of available sheet space while preserving appropriate spacing for readability. Linework should be concise, avoiding unnecessary extensions, and should employ straight lines with 90-degree angles wherever feasible.
Clear and detailed symbology, accompanied by precise annotations to provide equipment identification and applicable ratings, is required. The use of color coding shall be employed to distinguish between equipment installed on different feeders.
Diagrams must prioritize logical organization, comprehensive labeling, and the inclusion of a detailed legend to ensure they are professional, visually clean, and easy to interpret.

Source data and product data remain the property of the US Government. The Contractor may be required to explain and demonstrate the company's process for protecting all geospatial data, including but not limited to geometry, attributes, metadata, topologies, and relational database schemas and operations used in association with this PWS. The Contractor may be required to sign a non-disclosure agreement attesting to the same before source data are released. Further information about security and non-disclosure requirements should be obtained from the AFCEC GIO and/or installation GIO. Some installation map data, source and/or product, may be considered by the government to be “sensitive, but unclassified” or “Critical Unclassified Information (CUI)”. The intent of this clause is to prevent intentional or unintentional dissemination of “sensitive, but unclassified” information to include unauthorized access to the source and product data by any entity wishing to do harm to the USAF or US Government while the data resides on the Contractor's computer network. The Contractor is not authorized to release this information to any third party without the explicit consent of the Air Force Civil Engineer Center (AFCEC) or the involved installation. All source information must be returned to the government POC or destroyed upon completion of the project. Special requirements for handling classified map data, if applicable, will be addressed elsewhere in this PWS. Final deliverable geospatial data must be provided to the AFCEC GIO and installation GIO in an electronic, digital format according to specifications provided above.

3.2.6.2 **Geospacial Data Required Attributes**

The GIS information required to be provided but not limited to is:

Transformers(sdsID):

- Transformer ID
- End of line arrestors (Y/N)
- Fault Detector(Y/N)
- Rated Size (kVA)
- Year of Manufacture
- Serial Number:
- Fused:
- Fuse Size (if applicable)
- Padmount/polemount:
- Year of installation(utilityNetworkSubtype)
- Material Type (generalMaterialType)
- Condition Rating (condRatingValue)
- Operational Status (operationalStatus)
- Location Coordinates (lattitudeFrom &to, LongitudeFrom & to).
- RPUID (realPropertyUniqueIdentifier)
- Fed from:
- Looped: y/n
- Looped to feeder:
- Generator downstream:

Substations(sdsID):

- Substation Name
- Rated Size
- Year of Manufacture
- Serial Number
- Year of installation(utilityNetworkSubtype)
- Material Type (generalMaterialType)
- Size(length)(lengthSizeUOM)
- Condition Rating (condRatingValue)
- Operational Status (operationalStatus)
- Location Coordinates (lattitudeFrom &to, LongitudeFrom & to).
- Material Type (generalMaterialType)
- RPUID (realPropertyUniqueIdentifier)
- Any Single point of failure present:

Switch(sdsID):

- Switch ID
- Arrestors
- Model name
- Model Number
- Serial Number
- Manufacturer date
- Padmount:
- Install Date (installedDate)
- Location Coordinates (lattitudeFrom & to, longitudeFrom & to)
- Condition Rating (condRatingValue)
- Operational Status(operationalStatus)
- RPUID (realPropertyUniqueIdentifier)

- Crossfeed:
- Crossfeed feeder:

Feeder(sdsID):
- Feeder Capacity Amps
- Feeder cable size
- Feeder Cross Tie (at locations or switches). Y/N -1
- Feeder (Name):
- Year of Installation – (installedDate)
- Location Coordinates (lattitudeFrom & to, longitudeFrom & to)
- Condition Rating (condRatingValue)
- Operational Status(operationalStatus)
- Size(length)(lengthSizeUOM)
- Total Length (measured Length)
- RPUID (realPropertyUniqueIdentifier)

Cable(sdsID):
- Date of manufacturer
- Cable size (diameter)
- Total Length (measuredLength)
- Year of Installation – (installedDate)
- Location Coordinates (lattitudeFrom & to, longitudeFrom & to)
- Condition Rating (condRatingValue)
- Operational Status(operationalStatus)
- Size(length)(lengthSizeUOM)
- RPUID (realPropertyUniqueIdentifier)

The SDSFIE requirement is presented above as ATRIBUTE NAME. Failure to follow the maximum character limit will cause the characters to be truncated when uploaded in to the GIS system. All SDSFIE use (utilityNetworksubtype) as Electrical. Coordinate with the RealProperty to get realPropertyUniqueIdentifier properties and if not they will be left blank.

Cable routing shall be placed on the GIS map and will be an approximate location and not an actual routing diagram unless the shop concurs with the routing of the cables.

All attributes will also be included in a Microsoft Access or compatible database in addition to the GIS file as defined in 3.2.6.1.

3.2.7 Coordination of Outages

Following contract award the Contractor shall coordinate with the BCE to identify any areas of concern within the existing distribution system. The Contractor shall also work with the BCE to determine which distribution equipment and facility service disconnects OCPD are to be field verified using outages. Data collection shall be performed using energized work permits, refer to Section 3.2.10, however if energized work is not possible, outages shall be coordinated with the BCE.

The Contractor shall develop a list of all proposed outages in coordination with the BCE. The proposed outage schedule shall include the number and type of equipment/information anticipated to be verified during each outage. The Contractor shall submit the proposed outage schedule to the BCE and COR for evaluation and approval. The BCE and COR may reject any proposed outage schedule and/or provide recommended alternatives. The Contractor shall be required to receive signatures on the approved outage schedule from both the BCE and COR and shall retain a copy of the signed outage schedule. The Contractor shall resubmit the outage schedule for approval and signatures whenever a change is made to the proposed outage schedule. All submitted changes to the proposed outage schedule shall include a narrative describing the reason for the change. All proposed power outages shall be coordinated and submitted to the BCE and COR for approval and signed a minimum of 15 working days in advance of the outage. Additional time may be required for outage coordination depending on the outage policies at each installation, refer to Attachment 1 - Base Specific Requirements.

The Contractor shall ensure that no outage affects more than 10 facilities at any one time.

To the greatest extent possible, the Contractor shall schedule all outages during base/facility non-working hours, unless otherwise required by the installation's outage policy, refer Attachment 1 – Base Specific Requirements.

Following each field data gathering site visit, the Contractor shall submit a copy of the outage schedule to the COR indicating which outages had taken place during the site visit.

3.2.8 Energized Work Permits

Following contract award the Contractor shall coordinate with the BCE, a minimum of 15 working days in advance of any site visits, to identify any areas of concern within the existing distribution system. The Contractor shall also work with the BCE to determine which distribution equipment and facility service disconnects OCPD will require energized work permits to be field verified.

The Contractor shall develop all required energized work permit requests in coordination with the BCE. All energized work shall be performed and requested in accordance with UFC 3-560-01, NFPA 70E, AFMAN 32-1065, and local base policies. The Contractor shall submit the energized work permit requests to the BCE for evaluation and approval. The BCE may reject any energized work permit request and/or provide recommended alternatives. The Contractor shall be required to receive signatures on the approved energized work permit from the BCE and shall retain a copy of all approved energized work permits. The Contractor shall submit new/revised energized work permit requests for approval and signatures whenever a change is made to the work to be performed at each piece of equipment.

The Contractor shall be responsible for providing all necessary PPE. Contractor personnel shall wear all necessary PPE in accordance with the manufacturer's instructions during all energized work.

Following each field data gathering site visit, the Contractor shall submit a list to the COR indicating where energized work had been performed during the site visit.

3.2.9 Safety

The Contractor shall be required to wear all necessary PPE, as dictated by NFPA 70E, UFC 3-560-01, and IEEE C2, at all times when performing field data gathering activities. The Contractor shall also be required to close all distribution equipment enclosures, panels, and manholes and return equipment to a safe condition following field data gathering activities. If the Contractor discovers any existing safety concerns during field data gathering, such as open equipment enclosures or manholes, the Contractor shall notify the government, refer to Section 3.2.5. The Contractor shall be allowed no more than two (2) safety violations, such as improper use of PPE or leaving equipment in an unsecured or unsafe condition, per installation.

The Contractor shall provide personnel who have received appropriate training/education for the work to be performed.

3.3 BUILD/UPDATE MODEL (CDRL A001)

The Contractor shall be required to update the existing distribution system model based on information gathered during the preliminary and field data gathering phases. If an existing model does not exist, then the Contractor shall instead be required to create a new model for the distribution system.

The distribution system model shall be laid out according to the approximate geographic location of the actual equipment to allow the BCE to more easily locate equipment/devices needing to be updated in the future. If a model already exists, the Contractor shall verify that all model components match the information gathered in the field. If an error or conflict is found between the existing model and information gathered in the field, the Contractor shall make the necessary corrections. All devices containing assumed values within the model shall be clearly indicated as such through the use of specific colors, line types, and/or notes.

All distribution system devices and equipment shall be included in the model. These devices and equipment include items such as facility main service disconnects, transformers, primary power generation equipment, switches, sectionalizing cabinets, cables, manholes, poles, distribution cables, relays, CTs, breakers, and fuses. All breakers and relay settings shall be incorporated in the distribution model. The Contractor shall be responsible for modeling any devices or equipment which are not included in the standard software libraries. Following completion of each study, any custom library items created by the Contractor during the course of creating or updating the software model should be forwarded to the software manufacturer for incorporation into their standard library database. If an existing model exists and is found to be missing any devices, then the Contractor shall be responsible for adding the devices to the model. The distribution system model shall start at the output of the utility transmission lines.

If the BCE has indicated that the normal distribution system configuration routinely changes due to normal base operations, seasonal changes, routine maintenance, or any other factor other than emergency maintenance then the model shall include ‘scenarios’ to reflect each of the normal configurations. The system analyses listed below shall be computed for each scenario and the worst-case values included in the report. In addition to the worst-case values, the report shall also indicate which scenario resulted in the worst-case value for each piece of equipment.

3.4 SYSTEM ANALYSIS (CDRL A002A, CDRL A002B)

All portions of the distribution system (both single-phase and three-phase) shall be included in the System Analysis. Refer to Section 3.3 for additional information regarding worst case values.

3.4.1 Power Flow Analysis

The power flow analysis shall use existing peak and projected peak power demands, as determined by the 5-Year Plan, under various operating modes. The Contractor shall include motor loads of 50HP or more in the power flow analysis. Using the power flow analysis, the steady state active and reactive power flows shall be determined for system branches and buses, as well as bus voltage and power factor. As part of the Draft Report and Final Report, the impact of the existing and projected loads on the electrical system and equipment shall be reviewed, including factors such as:

- MW and MVAR flow from the utility service
- Voltage (Provide a separate list of all buses deviating from the acceptable voltage range, as determined by ANSI C84.1 Table 1 ‘Voltage Range A’)
- Power factor (Provide a separate list of all transformers having a power factor lower than 0.90)
- Total Harmonic Distortion (Provide a separate list of all transformers having a Voltage THD of 5% or more and/or a Current THD of 20% or more)
- MW and MVAR in each feeder and transformer
- Transformer tap settings
- Relay settings at the substations

- Fuse sizing for feeder branches
- Feeder conductor sizing
- Capacity of substations, switching stations, and primary power generation
- Paralleling the feeders and/or substations
- Ampere flow on each line section (Provide a separate list of all line sections over 90% loaded)

The power flow analysis shall be used to determine whether all components of the existing distribution system are adequately rated and have the proper OCPD ratings/settings for the existing and projected peak power demands. The analysis shall also be used to identify any potential power quality issues at any point in the distribution system because of excessive THD, poor PF, or excessive voltage drop. Power quality values are to be analyzed using both field measurements and model calculated values. When either are not acceptable, they are to be reported.

If any deficiencies are identified in the power flow analysis, Contractor shall provide recommendations for corrective actions and place reasonably grouped projects in Attachment xx – Project Recommendations. While the top five (5) shall be included in the executive summary.

The Contractor shall identify any deficiency or limitation which could impact the distribution systems ability to accommodate the 5-Year Plan. The Contractor shall make recommendations regarding additional or modified equipment or circuitry necessary to accommodate the 5-Year Plan. The additional equipment shall be coordinated with the present system such that:

- All projected loads shall be adequately served by the system, with minor extensions to be included with each construction project.
- The voltage drop in any part of the base primary system shall not deviate from the acceptable voltage range, as determined by ANSI C84.1 Table 1 ‘Voltage Range A, when all projected loads are added.
- Sufficient sectionalizing switches shall be provided for maintenance without loss of large segments of loads. Gang operated load break switches on overhead system and pad mounted switchgear following the base design standard on the underground system shall be proposed.
- The entire system shall possess the capability of safe break through the use of pad mounted switches and disconnect switches at any point on the system under load conditions.
- The system power factor shall be 90 percent or better. Capacitor power factor correction shall be justified, if proposed.
- Underground construction shall be considered in all areas. Recommendations shall incorporate a looped concept. If there are any single transformers radially feed, list as a deficiency and recommend areas where there could be a looped system for the transformer.
- The long-range goal is to convert the entire system to a primarily underground system.
- All proposed equipment shall be properly coordinated as to proper fault ratings and load settings with existing system, utility system and alternate source systems.
- Balanced feeder loading, outage localization, and reroute capability to maintain power to mission essential and critical loads by loop feed shall be included in the Contractor’s recommendations.

3.4.1.1 Motor Starting Analysis

A component of the power flow analysis shall be to evaluate the impact to the distribution system resulting from motor starting currents. The motor starting analysis shall evaluate the impact of individual motor starting currents. When performing the motor starting analysis, the Contractor shall identify any OCPD which could trip as a result of motor starting current. The Contractor shall also identify any potential power quality issues (such as excessive voltage drop) which result from the motor starting analysis and include the top five(5) recommendations in the executive summary.

3.4.1.2 Cold Load Inrush Analysis

The power flow analysis shall also evaluate whether the existing system is able to accommodate the cold load inrush that would be anticipated when reenergizing a feeder after a prolonged outage. The cold load pickup shall include inrush currents from all service transformers and motors rated 50HP or more. The Contractor may omit a motor from the cold load inrush evaluation if controls are confirmed and documented to be in place which would prevent that specific motor from activating following a power outage.

3.4.1.3 Cross Connect Capability

A component of the power flow analysis shall be to evaluate the cross connect capabilities of the existing system. Cross connect capability is the ability of one or more distribution feeders to be sub-fed from any other feeders using devices such as sectionalizing switches to accommodate occurrences such as maintenance without resulting in widespread outages.

The cross connect evaluation shall determine whether the existing distribution system has adequate cross connect capability to allow any distribution feeder to be sub-fed from any other feeder. The Contractor shall identify any single points of failure where a fault or maintenance operations could not be isolated without resulting in widespread outages. The Contractor shall also identify whether any feeders would be unable to be sub-fed from any other individual feeder without incurring some limitations on the distribution system (*for example if a feeder was unable to be sub-fed from any other feeder during peak loading*). For each feeder, the Contractor shall attempt to identify a minimum of one viable cross connect scenario that would allow the feeder's full peak load to be sub-fed from another feeder without limitation.

The Contractor shall evaluate the feeder’s cross connection for mission critical facilities only that have been identified in the base’s Emergency 702 plan.

The Contractor shall evaluate each feeder’s ability to be sub-fed from another feeder but is not required to separately evaluate every possible cross connect configuration. The Contractor will attempt to find at least one viable cross connect configuration for each feeder. Any feeder that is not identified with a viable cross connection is to be identified as a deficiency and be added in the recommendations. For this study, adequate cross connect capability shall be defined as the ability of each distribution feeder to be completely sub-fed from one or more other feeders.

If any deficiencies are identified while evaluating cross connect capability, or if a viable cross connection cannot be found for one or more feeders, Contractor shall provide recommendations for corrective actions.

3.4.1.4 Distribution Reliability Study

A component of the power flow analysis shall be to perform a distribution reliability study for each installation. The distribution reliability study shall be used to determine the baseline reliability of each feeder.

The distribution reliability study shall be comprised of two components. The first is the completion of a reliability study, performed in accordance with IEEE 493, to determine the reliability of each distribution feeder. The second component shall be the recommendation of changes aimed at improving the reliability of the distribution system to a minimum of 99.9 to any load.

During the field data gathering site visits, the Contractor shall document system vulnerabilities which could impact reliability such as poor equipment condition and single points of failure. Based on the vulnerabilities identified, the Contractor shall make recommendations such as:

- Repairing/replacing equipment
- Installing new switches and cross connect points
- Eliminating single point of failures (SPOF) to all generator locations.
- Providing back feed capability to an alternate path of power to the primary of thee service transformer with a loop configuration
- Recommend cross connections to feeders serving multiple generators

Within the body of the report the Contractor shall list: Inherent Availability (Ai) defined in IEEE 493 and Unavailability (hrs/year) defined in IEEE 493 at each substation as well as the entire distribution system. The Contractor shall provide a statement based on the Inherent Availability (Ai) and IEEE 493 on the adequacy of the overall current system reliability. The Contractor shall include the entire reliability report as an attachment to the report.

The primarily list of findings must be submitted to the AFCEC SME for approval upon completion of the analysis, AFCEC SME will select the top 5. For each recommended change, the Contractor shall provide a quantified value for the anticipated improvement to the system's/equipment's reliability. A table compiling all system vulnerabilities identified for each building with a generator, the Contractor's recommended corrections, and the quantified improvement to reliability shall be included in the Appendices. With the top five (5) projects to increase the IEEE 493 number for the critical facilities be included in the executive summary.

3.4.2 Short Circuit Study

The Contractor shall perform short circuit analysis using the software chosen for each location to determine if existing equipment is adequately rated to accommodate available fault currents resulting from the existing and planned utility system service. The short circuit studies shall include fault current contributions from the incoming utility service, through the pertinent high and medium voltage buses, through the secondary side of the utilization transformers and down to the building service entrance panels of the buildings and other facilities.

If any devices are found to have insufficient short circuit ratings, the Contractor shall make recommendations on corrective actions *(such as replacing the device)*. Contractor shall list all over duty equipment in the body of the report with the to five (5) listed in the executive summary

3.4.3 Protective Coordination Study

The Contractor shall show protective device coordination using the software chosen for each location, to include optimizing the coordination settings. The Contractor shall request that the BCE provide all existing protective device settings but shall validate data during field gathering.

The Contractor shall use the software to generate time current curves (TCC) to verify protective device coordination for all existing OCPDs under normal operating conditions. Using the software generated TCCs, the Contractor shall identify locations where the existing OCPSs are not coordinated or are only partially coordinated. The Contractor shall then provide recommendations to improve the protective device coordination.

After determining which changes are recommended to improve the system coordination, the Contractor shall create new TCCs incorporating the recommendations. Wherever a recommendation is made to improve coordination, the existing and recommended TCCs shall be included in the Report to illustrate the improved coordination. In addition, TCCs for all critical and mission essential infrastructure and equipment shall be included in the Report to verify proper coordination regardless of whether recommendations are made. Incorporate this in an excel spreadsheet and be included in the submittals. (Should we have this in an excel format so it is easier to search? Something like "the recommendations should be included in the report and in an excel spreadsheet as an attachment.)

3.4.4 Arc Flash Hazard Analysis

The Contractor shall perform an arc flash hazard analysis to determine incident energy, arc flash protection boundaries, working distance, and required PPE for all equipment on the primary electrical distribution system using the software chosen. The report shall provide the worst-case arc hazard values for each equipment location. Calculations shall be performed to determine the above listed values for both the line side *(immediately upstream of the main OCPD)* and load side *(immediately downstream of the main OCPD)* of equipment provided with OCPDs. Requirements of NFPA 70E shall be complied with.

All equipment within the range of 208-15kVac shall be analyzed using the latest edition of IEEE 1584 calculation method. While any other voltages above 15kV or outside of the range of IEEE 1584, shall use either IEEE C2 or NFPA 70E based calculations.

If any medium voltage bus is found to have an incident energy of 12 cal/cm^2 or greater, the Contractor shall provide recommendations for changes (*such as changing OCPD settings or replacing fuses*) to reduce the incident energy below 12 cal/cm^2. If any low voltage bus is found to have an incident energy of 40 cal/cm^2 or greater, the Contractor shall provide recommendations for changes to reduce the incident energy below 40 cal/cm^2. Whenever a bus having an incident energy equal to or greater than the values indicated above is unable to be reduced below the indicated value, an explanation shall be provided indicating any reasons why. The Contractor shall provide a list, within the body of the report, of all devices with calculated incident energies greater than the listed thresholds above.

After determining which changes are recommended to improve the system arc flash values, the Contractor shall create new TCCs incorporating all recommendations to illustrate any impact on coordination, the existing and recommended TCCs shall be included in the Report.

3.4.5 Deficiencies and Recommended Projects

The Contractor shall use the system analysis to evaluate the existing distribution system for deficiencies. Any deficiencies identified in any of the above listed components of the System Analysis, including recommended OCPD changes, shall be included in the report along with recommended corrective actions. If no recommended changes or corrective actions are provided for any individual component of the System Analysis, the Contractor shall indicate this in the report and provide a detailed explanation of why no changes are recommended. Additionally, the Contractor shall provide recommended corrective actions to correct all known issues and primary concerns for each installation as well as any other requested recommendations listed in Attachment 1 – Base Specific Requirements. For each recommendation, the Contractor shall perform a cost estimate which accounts for geographic cost factors such as shipping costs, local utility charges, etc.

The Contractor shall consider the potential impact of any current or future projects identified by the 5-year plan when evaluating deficiencies.

All projects shall follow Attachment 4 – Project Recommendations spreadsheet and be a deliverable in excel format.

3.4.6 Base Capacity Assessments

The Contractor shall be provided electrical peak demand and/verses capacity for the base primary and secondary substations to ensure concurrence maintenance can be performed to prevent loss of power to facilities during the peak demand months.

The Contractor shall determine if the base demand exceeds capacity for concurrent maintenance at primary and secondary substations. If peak demand is within 10% of the system capacity, make recommendations for improvement. Provide complete analysis of primary and secondary substation peak demand and capacity. The analysis must include future proposed projects and existing projects impacts.

Reference site specific Attachment 1 – Base Specific Information for additional loads or distributed energy sources that may need to be considering in the system evaluation for electrification.

3.4.7 Electrical Generator Inventory and Survey

The Contractor shall use APIMS data provided by the government in Attachment 3 – APIMS Data to account for generators in the models. Populate an inventory list to be concurred by the base in Attachment 5 – Generator List. The base will need to agree with the list of generators on site. The Contractor will need to get name plate data and verify configuration for both the generator and the associated ATS or MTS. If there is any discrepancy between Attachment 5 – Generator List and the base the Contractor shall document changes and be submitted to AFCEC/COSM.

3.5 DRAFT REPORT (CDRL A002A, CDRL A002B)

Upon completion of the System Analysis, the Contractor shall develop a Draft Report. The Draft report shall include the completed results of all analyses described within Section 3.4. The Contractor shall attempt to verify and correct all discrepancies between the report and appendices before submitting to the government for review. The Contractor shall submit in addition to the report, the software model, pdf versions of distribution map and one-line. The Draft Report shall be submitted to the AFCEC COR, AFCEC PE, USACE POC, installation POC, Base Project Manager, and the COMMAND for review and comment. The Contractor shall allow four (4) weeks for review. The Contractor shall host a teleconference within one (1) week following the Draft Report Review period to discuss comments and concerns from all stakeholders. Following the Draft Report review teleconference, the Contractor shall be allowed four (4) weeks to incorporate comments and make updates.

3.5.1 Documentation of Assumptions

The Contractor shall individually document all assumptions, made for any reason, within the body of the report. At a minimum this documentation shall include a detailed description of the assumption made including actual values or equipment assumed, reasons for making the assumptions *(such as missing nameplate information)*, a narrative describing the appropriateness of the assumption, and the name, title, and organization of the personnel coordinated with when making the assumptions (refer to Section 3.2.4). A general description of an assumption which applies to multiple devices (but not necessarily all devices of that type) is acceptable if all the documentation required above is provided, each device that the assumption applies to is clearly listed, and the exact same assumption was made for each device for the exact same reason. General descriptions of typical locations where any assumption may be made will not be considered sufficient documentation.

Equipment ID	Assumption	Reason for Assumption	Justification	Coordinating Individual

T0001	Z%=4.5%	Missing Nameplate	IEEE C57.12.XX indicates that this is a typical value for this type and size of transformer	Jane Doe / Electrical Engineer / 11 CES
T0002	Z%=4.5%	Impedance Missing from Nameplate	IEEE C57.12.XX indicates that this is a typical value for this type and size of transformer	Jane Doe / Electrical Engineer / 11 CES
Fuses: F0001, F0002, F0003, and F0004	Fuse Type: Dual Sensing C10	Bay-O-Net Fuse data not available	Bay-O-Net fuse assumed based on manufacturer literature and recommendation	Jane Doe / Electrical Engineer / 11 CES
Fuses: F0005, F0006, F0007, and F0008	Fuse Type: Dual Sensing C12	Bay-O-Net Fuse data not available	Bay-O-Net fuse assumed based on manufacturer literature and recommendation	Jane Doe / Electrical Engineer / 11 CES

Example: Documentation of Assumptions Table

3.5.2 Additional Documentation in Appendices

In addition to the documentation related to the System Analysis, the following items shall also be included within the Appendices of the report.

3.5.2.1 Outage Verification Summary Table

The Contractor shall provide a table in the Appendices listing all equipment and information which was verified during the planned outage(s). For each piece of equipment that information was collected using an outage, the table shall include the following:

- Outage number *(see below)*
- Date of the outage
- Outage start time
- Outage end time
- Outage duration
- Equipment ID number
- List of all information verified on the equipment which required an outage *(see below)*

Each outage shall be assigned a unique identifying number. A single outage may be used to verify information from multiple pieces of equipment.

A simple description of the information verified with the outage is all that is required. The actual values recorded shall not be included in this table; for example, the Contractor would indicate "Short Circuit Rating" in the table but would not list the actual rating recorded. Refer to the Example Outage Verification Summary Table Below.

Outage Number	Outage Date	Outage Start Time	Outage End Time	Outage Duration	Equipment ID	Data Collected
001	2018-03-17	0800	1030	2.5 Hours	T0001	• Fuse Data
					SW0001	• Fuse Data
002	2018-03-17	1100	1130	0.5 Hours	B0001	• Main OCPD Settings • Main Disconnect Short Circuit Rating

Example: Outage Verification Summary Table

3.5.2.2 Energized Work Summary Table

The Contractor shall provide a table in the Appendices listing all equipment and information which was verified using energized work permits. For each piece of equipment that information was collected using energized work, the table shall include the following:

- Energized work permit number

- Date of the energized work
- Energized work start time
- Energized work end time
- Equipment ID number
- List of all information verified on the equipment which required energized work *(see below)*

A simple description of the information verified during the energized work is all that is required. The actual values recorded shall not be included in this table; for example, the Contractor would indicate "Short Circuit Rating" in the table but would not list the actual rating recorded. Refer to the Example Energized Work Summary Table Below.

Energized Work Permit Number	Date	Energized Work Start Time	Energized Work End Time	Equipment ID	Data Collected
001	2018-03-17	0800	1030	T0001	• Fuse Data
002	2018-03-17	1100	1130	B0001	• Main OCPD Settings • Main Disconnect Short Circuit Rating

<u>Example: Energized Work Summary Table</u>

3.5.2.3 Compiled OCPD Information Table

The Contractor shall provide a table in the Appendices which compiles information for each existing OCPD installed in the distribution system. The government's intent is for the Compiled OCPD Information Table to be used in the future by electricians to implement recommended changes. At a minimum, this table shall include the following information for each OCPD:

- Location
- Equipment ID Number
- Device Number/Name
- Open/Close Status
- Coordination Status
- Voltage
- CT Ratio
- Manufacturer
- Type
- Rating
- Settings
- Recommended Change (Y/N)
  (If yes, include reference to location of recommendation. If setting changes are recommended, include the recommended setting changes in the table as well.)

Location	Equipment ID	Device Name/ Number	Open/ Closed Status	Coord. Status	Voltage	CT Ratio	Manuf.	Type	Rating	Settings	Rec. Change (Y/N)
Sub. #1	Sub. #1 Main 1	Circ. 1 Relay	Closed	Partially Coord-inated	12.47KV	200:5	ABB	DPU-2000	300A	Tap:7.5 Time Dial:1.5	Y (Refer to Sec. 6.E)

<u>Example: Compiled OCPD Information Table</u>

3.5.2.4 Summary of Key Points

The Contractor shall provide a table in the Appendices which compiles information for the base distribution system being studied in its existing condition. This table shall include the following distribution system information:

- Base Name
- Number of Substations
- Substation Identifier(s)
- Substation Name(s)
- Number of Distribution Feeders
- Distribution Feeder Identifiers
- Distribution Feeder Names
- Distribution Feeder Owner(s)

- Substation Owner(s)
- Substation Transformer Size(s) (KVA)
- Number of Switching Stations
- Switching Station Identifier(s)
- Switching Station Name(s)
- Switching Station Owner(s)
- Software Start and End Date
- Software License Number
- Distribution Voltage(s)
- Installation Peak Demand (KVA)
- Percent Overhead Distribution
- Percent Underground Distribution
- Cross-Tie Capability Adequate? (Y/N)
- Number of Service Transformers Per Feeder
- Software POC, Name, e-mail, address, and phone number

If any of the above listed items (such as substations or switching stations) do not exist at an installation, it shall be indicated on the Installation Key Points Summary. If ownership of any substations, switching stations, or feeder is shared between more than one organization or entity, the Installation Key Points Summary shall state this and indicate the demarcation points between organizational ownership. The Attachment 7 – Key Points Summary shall be in addition to the including above information for the report and be provided in a separate document for each base and be in Key Points 2 tab. (*Example: Main Substation is shared ownership between the Utility Company and Installation. Utility owns all equipment upstream and including the substation transformer secondary lugs. Installation owns all equipment downstream of the substation transformer secondary lugs.*)

Attachment 7 – Key Points Summary, Key Points 1, 2, and 3 shall be filled out with data. Key Points 1 is to include information about the number of devices and equipment, either deficient or total counts as described. Key Points 3 is to provide feeder and substation transformer load information.

3.6 FOLLOW-UP DATA GATHERING

Following the Draft Report review teleconference, the Contractor shall coordinate with the installation BCE to schedule any additional field data gathering site visits needed, including any additional power outages and/or energized work permits that might be necessary. The Contractor shall use these follow-up data gathering site visits to collect any remaining information needed to address comments from the Draft Report Review and complete the System Analysis in preparation for developing the Final Report.

Follow-up data gathering site visits shall follow the same procedures as the initial field data gathering site visits described in Section 3.2.

3.7 FINAL REPORT (CDRL A002A, CDRL A002B)

All comments received during the Draft Report Review shall be addressed and incorporated into the Final Report. The outline for the Final Report shall be identical to the Draft Report. All report submission and review requirements shall be identical to that required for the Draft Report. The data entered into the software model submitted with the Final Report shall have no more than two (2) discrepancies when compared to actual distribution equipment data. Within one (1) week following the Final Report review period the Contractor shall host a review teleconference identical to the meeting held for the Draft Report Review.

3.8 CORRECTED FINAL REPORT (CDRL A002A, CDRL A002B, CDRL A007)

All comments received during the Final Report Review shall be addressed and incorporated into the Corrected Final Report. The outline for the Corrected Final Report shall be identical to the Draft Report. All report submission requirements shall be identical to that required for the Draft Report, except as modified below.

In addition to the submittals indicated above, the Contractor shall submit one (1) additional electronic media copy of the Corrected Final Report to the CO. All Corrected Final electronic media submittals shall be on CD(s) or DVD(s).

In addition to the electronic submittals required above, the Contractor shall provide two (2) tabbed paper copies of the report and all appendices in 3-Ring binders to the Base Project Manager, four (4) copies to the installation POC and one (1) copy to the COMMAND.

In addition to the submission requirements listed above, the Contractor shall also provide one (1), approximately 72” x 96” or larger, base electrical distribution map and one (1), approximately 72” x 96”, electrical one-line diagram map. Drawing size may be accomplished by (tapped) smaller plotted sections. If the 72” x 96” or larger drawing size is accomplished by (tapped) smaller plotted section, the smaller plotted sections shall be no smaller than 11” x 17”. The one-line diagram shall be created using AutoCAD (version to be coordinated with the installation). One-line diagrams shall reflect the approximate geographical arrangement of equipment and be designed for seamless assembly into a consolidated 72” x 96” layout by printing and taping individual sheets. To ensure clarity and consistency, all labels, linework, and symbols must conform to AutoCAD standardized conventions and symbols, maintaining uniformity in font, size, and spacing. All symbols and text shall be clearly legible without magnification. One-line diagrams should optimize the use of available sheet space while preserving appropriate spacing for readability. Linework should be concise, avoiding unnecessary extensions, and should employ straight lines with 90-degree angles wherever feasible. Clear and detailed symbology, accompanied by precise annotations to provide equipment identification and applicable ratings, is required. The use of color coding shall be employed to distinguish between equipment installed on different feeders.

Diagrams must prioritize logical organization, comprehensive labeling, and the inclusion of a detailed legend to ensure they are professional, visually clean, and easy to interpret. In addition to the hard copy indicated above, the Contractor shall also provide the AutoCAD and .pdf versions of the distribution maps and one-line diagram. All one-line diagrams and electrical distribution maps provided shall be fully up to date and accurately reflect the information collected during the data gathering process.

All Corrected Final documents shall be delivered in a single package to each individual indicated above regardless of if the Government has provided comments or not in a previous submittal.

The Contractor shall also provide all additional base specific submittals listed in Attachment 1 – Base Specific Requirements.

Within six (6) weeks following the Corrected Final Report submittal distribution, the Contractor shall provide a presentation to Civil Engineering government personnel to present, discuss and answer questions about the Corrected Final Report. The Contractor shall elaborate on any significant aspects or parts of the Corrected Final Report and its findings. The presentation will take place at each location, with the BCE office. The Contractor shall coordinate the date and location of the presentation with the AFCEC PE, USACE POC, installation POC, and Base Project Manager a minimum of four (4) weeks in advance for CONUS installations and a minimum of six (6) weeks in advance for OCONUS installations.

3.9 EQUIPMENT LABELING

3.9.1 Arc Flash Labeling

The Contractor shall provide arc flash labeling on all switchgear located within each substation and switching station. The Contractor shall also provide and install arc flash labeling suitable for exterior applications on all exterior power distribution equipment. The exterior equipment to be labeled includes distribution transformers (three-phase and single-phase), sectionalizing cubicles, and pad-mounted switchgear.

Where distribution equipment has documented routine maintenance complying with NEMA MTS or

NFPA 70B, the Contractor shall apply a Detailed Arc Flash Warning Label, in accordance with UFC 3-560-01 Figure 1-7. On equipment which the BCE has confirmed does not receive routine maintenance, the Contractor shall apply a red “DANGER” Arc Flash Label, in accordance with UFC 3-560-01 Figure 1-10.

For any equipment which is calculated to have an incident energy greater than 40 cal/cm^2, regardless of whether or not it receives routine maintenance, the Contractor shall apply a red “DANGER” Arc Flash Label, in accordance with UFC 3-560-01 Figure 1-11. Detailed Arc Flash Labels, complying with UFC 3-560-01 Figure 1-7, shall not be installed on equipment receiving a red “DANGER” Arc Flash Label, regardless of whether or not the equipment received routine maintenance.

Where a piece of equipment is protected by an upstream OCPD having a “maintenance” setting which modifies the trip characteristics of the OCPD to reduce the trip time below normal levels, if the upstream OCPD has documented routine maintenance complying with NEMA MTS or NFPA 70B, the Contractor shall apply a second Detailed Arc Flash Label for this lower “maintenance” level. It shall be clearly and conspicuously indicated on the “maintenance” Detailed Arc Flash Label that the values listed on it apply only when the upstream OCPD is placed in the “maintenance” mode.

The Contractor shall not apply Arc Flash Labels until after submission of the Corrected Final Report.

3.9.2 Equipment Identification Labeling

If, for any reason, the equipment naming convention used in the distribution system model and report does not match the existing equipment identification labels the Contractor shall relabel the existing equipment with the equipment naming convention used in the distribution system model and report. Whenever a new naming convention is created for all or part of the existing distribution equipment, the Contractor shall also include a cross-reference guide in the report showing the correlation between the new and existing equipment IDs.

The Contractor shall also provide new identification labels for any existing pad mounted equipment found without an identification label during the field data gathering process.

3.10 SOFTWARE AND TRAINING (CDRL A001, CDRL A008, CDRL A009)

3.10.1 Software and Training Provided to Installations

The Contractor shall provide software and training for shop personnel on the use of the software prior to completion of the first model submission. As part of the training, base personnel will be trained to update the program as changes to the distribution system occur. During the training, the Contractor shall answer questions about the program and demonstrate examples of the proper way to utilize each component of the software.

The software license provided shall be the same software and version used by the Contractor to perform the System Analysis, shall include all applicable software modules (Such as: Power Flow, Short Circuit, Electrical Coordination, Arc Flash, Harmonics, Single Phase, Unbalanced, Reliability), GIS Map import and export capability, Digital twin with capability of integrating into the AMRS system, and shall have unlimited bus capability. Only 10software license(s) are to be provided and shall have a minimum of three (3) years maintenance/subscription remaining at the time the Corrected Final Report is submitted.

If the software used to perform the System Analysis at an installation is the same as the software used at the installation, then the Contractor shall provide the installation with an updated software license maintenance plan for their existing license, including all applicable software modules. The updated software license must have a minimum (3) three years of maintenance at the time the

Corrected Final Report is submitted. For estimating purposes, assume new license and three (3) years maintenance will need to be purchased to handle any delinquency of maintenance costs for the software license.

Unless specifically indicated otherwise in Attachment 1 - Base Specific Requirements, the software requirements listed above shall be applied uniformly to each installation.

Any new software licenses provided by the Contractor shall be a non-network license. The Contractor shall install the software on a Contractor furnished, standalone laptop computer having an operating system and performance specifications which are fully compatible with the software provided. The Contractor furnished, standalone computer shall become property of the government upon completion of the project. Software licenses requiring a USB key are not acceptable. Whenever a new software license is provided to an installation, the Contractor shall also provide that installation with one (1) standalone laptop computer installed with a complete set of the software licenses. The Contractor shall purchase 3 years' worth of maintenance or subscription for the bases shall be done at the beginning of the contract. This maintenance will be put into a pool of licenses the bases can access through a web portal.

If a stand alone computer is required for the location it will be a laptop that will have a minimum of a workstation with an i7 processor, 32GB of RAM and 2TB of storage with DVD-RW data reading and writing capability. These laptops will not be connected to Government systems and will not be required to be added in the future and shall be marked appropriately.

At each installation, regardless of the software provided, the Contractor shall be responsible for providing software training and training material for the number of individuals indicated in Attachment 1 – Base Specific Requirements, at a minimum. With the exception of any additional computers provided by the Contractor to facilitate training, all training material shall become property of the government following training and separate training material shall be provided for training at each installation. Training material shall include a training syllabus and tabbed paper-training documents in 3-Ring binders. A component of the software training shall consist of hands-on training with the software.

Uniformed personnel routinely rotate in and out of each installation. As part of the software training material, the Contractor shall provide detailed, written software instructions of sufficient clarity to allow future uniformed personnel to perform self-guided software training after completion of the contract.

The software training shall be a minimum of 16 hours.

In Addition to the additional license at least 10 more training seats should be negotiated with the software company that is chosen in order to help facilitate additional learning after on-site training or at the discression of AFCEC/COSM where the training seats should go.

3.10.2 Software and Training Provided to AFCEC

The Contractor shall provide AFCEC with two (2) standalone computers each loaded with non-network software licenses of each software (other than EasyPower) used to conduct the System Analysis prior to completing the first model submission. The software provided to AFCEC shall be identical to the software provided to the installations, refer to Section 3.10.1. The Contractor shall provide AFCEC or AFCEC's designated representative with updated three (3) year software license maintenance plan for each software used for two (2) of AFCEC's existing software licenses.

The Contractor shall also be responsible for providing software training and training material to four (4) individuals from AFCEC for each software used. The training and materials provided for AFCEC shall be identical to the training and materials provided to the installations, refer to Section 3.10.1, with the exception that any installation specific training may be replaced with more generalized training. Any software training and training material provided to AFCEC must be submitted at a minimum of one (1) month prior to using that software.

4 APPLICABLE CRITERIA

The Contractor shall comply with all applicable (1) federal, state, and local environmental statutes, instructions, manuals, handbooks, regulations, guidance, policy letters, and rules (including all changes and amendments) and (2) Presidential Executive Orders, in effect on the date of issuance of this PWS.

All work shall be in accordance with criteria contained in current United Facilities Criteria; Air Force programming, design, and maintenance guidance including Air Force instructions and manuals, and Air Force Engineering Technical Letters; national building construction codes; and industry standards (ANSI, IEEE, NEC, NESC and others). The most important technical references for this project include:

- ANSI C84.1-2020, Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz)
- IEEE 141-1993 (R 1999), Electric Power Distribution for Industrial Plants (IEEE Red Book)
- IEEE 241-1990 (R 1997), Electric Power Systems in Commercial Buildings (IEEE Gray Book)
- IEEE 242-2001, Protection and Coordination of Industrial and Commercial Power Systems (IEEE Buff Book)
- IEEE 399-1997, Industrial and Commercial Power System Analysis (IEEE Brown Book)
- IEEE 446-1995 (R 2000), Emergency and Standby Power Systems for Industrial and Commercial Applications (IEEE Orange Book)
- IEEE 493-2007, Design of Reliable Industrial and Commercial Power Systems (IEEE Gold Book)
- IEEE 519-2022, Harmonic Control in Electric Power Systems
- IEEE 1584-2018, Guide for Performing Arc-Flash Hazard Calculations
- IEEE C2-2023, National Electric Safety Code (NESC)
- IEEE C37.010-2016, Guide for AC High-Voltage Circuit Breakers > 1000 Vac Rated on a Symmetrical Current Basis
- IEEE C37.13-2015, low-Voltage AC Power Circuit Breakers Used in Enclosures
- IEEE C57.106-2015, Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment
- Unified Facilities Criteria (UFC) 3-501-01, Electrical Engineering
- Unified Facilities Criteria (UFC) 3-520-01, Interior Electrical Systems
- Unified Facilities Criteria (UFC) 3-540-01, Engine-Driven Generator Systems for Backup Power Applications
- Unified Facilities Criteria (UFC) 3-540-08, Utility-Scale Renewable Energy Systems
- Unified Facilities Criteria (UFC) 3-550-01, Exterior Electrical Power Distribution
- Unified Facilities Criteria (UFC) 3-560-01, Operation and Maintenance: Electrical Safety
- Linear Infrastructure Consolidate Playbook, 20 October 2014 or latest version published on the CE PORTAL.
- NETA MTS, Standard for Maintenance Testing Specifications
- NFPA 70, National Electrical Code
- NFPA 70B, Recommended Practice for Electrical Equipment Maintenance
- NFPA 70E, Standard for Electrical Safety in the Workplace
- AFMAN 32-1062 Electrical Systems, Power Plants and Generators
- AFMAN 32-1065 Grounding & Electrical Systems

Where conflicts exist between applicable criteria, the stricter criteria shall override.

5 SERVICE SUMMARY

Performance Objective (PO)	PWS Paragraph(s)	Performance Threshold	Method of Surveillance

PO1: Safety	PWS 3.2.9	Contractor shall have two (2) or fewer safety violations per installation.	Periodic Inspection (Monthly)
PO2: Perform Coordination and Field Data Gathering Activities	PWS 3.1 and 3.2	S: Plan, schedule, coordinate all items required to conduct data gathering activities and collect all information required during inspection with no more than three (3) discrepancies. Discrepancy is defined as failure to plan, schedule, and/or coordinate surveys within the Contractor approved PPC. Discrepancy is also defined as failure to collect all necessary information in accordance with this PWS.	Periodic Inspection (Semi-annually per base inspection)
PO3: Provide Corrected Final Reports and all Content in Appropriate Format	PWS 3.8	S: After Final submission comments are provided by the Government, successfully rectify all comments provided by COR with no more than one (1) submission of corrected final report.	100% Inspection (Per report)

6 PERSONNEL QUALIFICATIONS

The proposed Contractor personnel shall possess at a minimum the following:

- Knowledge of the National Electric Code (NEC) 70 and National Electric Safety Code
- Experience on projects, which are of similar scope, and include a minimum of 8 locations (in total) of similar size and complexity to the Air Force installations included in this contract, using the proposed software to perform electrical substation and distribution system modeling, power flow analysis, short circuit calculations, protective devices coordination, harmonic distortion analysis, power factor evaluation and arc flash calculation.
- Experience on projects, which are of similar scope, and include a minimum of 8 locations (in total) of similar size and complexity to the Air Force installations included in this contract, in the evaluation of the primary electrical system to establish future modifications, expansions, and upgrades to support future electrical requirements and improve system efficiency, reliability, and maintainability.
- All Contractor employees performing fieldwork are to be loaded into SPOT and submit for all applicable OCONUS locations for work to be performed. SPOT requests can be performed at https://spot.dmdc.mil

At least one personnel on site shall be an electrician with 3 years of experience and/or have NETA equivalent or greater testing certificates with equivalent practical experience.

The lead GIS/CAD technician shall have at least 10 years of professional industry experience completing GIS and/or CAD maps.

6.1 STATUS OF FORCES AND HOST NATION SUPPORT AGREEMENTS

The Contractor shall ensure employees are kept abreast of applicable laws, agreements, policies, etc. The Contractor shall comply with all applicable US and international law, Status of Forces

Agreements (SOFAs), and Host Nation Support Agreements (HNSAs). Unless addressed otherwise by international agreement, Contractor personnel are subject to the law of the nation in which they are located. The Contractor shall be prepared to comply with all local taxes, immigration requirements, customs formalities and duties, environmental rules, bond or insurance requirements, work permits, and transportation or safety codes. The fact that the supported Government entity may enjoy certain exemptions from local law does not mean Contractor personnel are also exempt. Generally, Contractor personnel are not exempt from local laws. Additionally, the Contractor is subject to host nation criminal law, unless specifically addressed otherwise by international agreement. Under United States law, Contractor employees and other civilians accompanying the armed forces may also be prosecuted by the United States Government for criminal acts. The Contractor is responsible for obtaining and complying with the most current copy of applicable agreements from the US Department of State official website.

7 ADMINISTRATIVE AND MANAGERIAL REQUIREMENTS

The Contractor shall provide management, planning, and performance measurement reporting pertinent to the performance of the requirements identified in this PWS.

7.1 MANAGEMENT, PLANNING, AND REPORTING SERVICES

The contractor shall attend and/or support meetings and teleconferences as required by the Contracting Officer (CO) and/or the Contracting Officer's Representative (COR). The purpose of the meetings includes contract discussions, progress reviews, project status, and feedback from the government on the review of draft submittals. The contractor will be responsible for drafting and submitting all meeting minutes to the CO and COR. The contractor will also be responsible for submitting any conference agenda(s) at least 24 hours prior to the conference.

The Contractor shall plan project activities, including the development, implementation, and maintenance of project schedules, events, status of resources, report(s) on the activities, and progress toward accomplishing project objectives. The Contractor shall document for government review and approval the results of the project efforts for this PWS.

a. Work Breakdown Structure (WBS) Requirements: Within 45 calendar days after task order award, the Contractor shall prepare and submit for approval a Work Breakdown Structure (WBS) to the CO, COR, AFCEC PM, AFCEC PE, USACE POC, Base Project Manager, and Installation POC. The WBS shall be used to report the schedule status for this project. (CDRL A004)

b. Schedule and Planning Requirements – Installation Project Planning Chart (PPC): The Contractor shall provide a Project Planning Chart (PPC) using Microsoft Project for tracking work progress at each installation through the use of a Gantt chart. The PPC shall be submitted monthly. The Contractor should consider the dates/timeframes on the PoP and Government Estimated performance schedule for each installation. (CDRL A006)

7.2 CONTRACTOR MANPOWER REPORTING

Service Contract Reporting (SCR) Requirements. In accordance with 10 United States Code (U.S.C.) 2330a, the DoD is required "to establish a data collection system to provide management information with regard to each purchase of services by a military department or Defense Agency for services." For DoD, this includes contract actions in specific service acquisition portfolio groups, to include logistics management services, equipment related services, knowledge-based services, and electronics and communications services that exceed $3M annually.

The Contractor shall report ALL Contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract via SAM. The Contractor is required to completely fill in all required data fields using the following web address: Systems Award Maintenance (SAM) https://www.sam.gov/SAM/.

Reporting inputs will be for the labor executed during the period of performance during each Government fiscal year (FY), which runs 1 October through 30 September. The reporting period for Contractors in SAM generally is open from mid-October to mid-December for reporting against the prior government fiscal year. SAM will only enable reporting for entities with contracts that meet the SCR thresholds. Contractors may direct questions to the SAM help desk.

Uses and Safeguarding of Information. Information from the secure web site is considered proprietary in nature when the task order number and Contractor identity are associated with the direct labor hours and direct labor dollars. At no time will any data be released to the public with the Contractor name and task order number associated with the data.

7.3 DATA PROTECTION AND DESTRUCTION

All digital files, final hard-copy products, source data acquired for this project, and related materials, including that furnished by the Government, shall become property of the Government. The Contractor will not issue, distribute, or publish these items without explicit Government permission. All documents (papers, reports, CD's, etc.) provided to the Government shall be marked CUI (Controlled Unclassified Information) and include a cover letter to be accepted by the Government. The Contractor must provide digital copies via CD ROM or other agreed upon digital media in their native format to the government and will return all prepared materials and Government-furnished documents to the COR before the Government can make final payment. Upon completion of the use of the data supplied pursuant to this agreement, the Contractor shall return all information certifying that all such data, any and all copies, reproductions, facsimiles, subsets, etc. have either been returned or destroyed. All records must be managed in accordance with AFI 33-322 Records Management.

The US Government information stored, collected, processed, and used on Contractor systems shall be protected in a manner that meets DoD and USAF standards for Controlled Unclassified Information (CUI).

At the conclusion of the period of performance, the Contractor will delete/remove all US Government information from Contractor owned computing systems and devices and provide a certification letter to the CO and COR notifying them the actions are completed to include the data, action taken for removal, and the signature of a company official.

8 STANDARD ATTACHMENTS

Supporting Documents:
Attachment 1 – Base Specific Requirements
Attachment 2 – Reserve
Attachment 3 – Reserve
Attachment 4 – Project Recommendation Guide
Attachment 5 – Reserve
Attachment 6 – AFMAN 32-1062 ELECTRICAL SYSTEMS, POWER PLANTS & GENERATORS
Attachment 7 – Key Points Summary
Attachment 8 – Reserved
Attachment 9 – Reserved

Attachment 10A – Arnold AFB – Distribution Map
Attachment 10B – Arnold AFB – One Line
Attachment 10C – Arnold AFB – Model
Attachment 10D – Arnold AFB – Study
Attachment 10E – Arnold AFB – Generator

Attachment 11A – Reserve
Attachment 11B – Reserve
Attachment 11C – Reserve
Attachment 11D – Reserve
Attachment 11E – Reserve

Attachment 12A – Nellis AFB – Distribution Map
Attachment 12B – Nellis AFB – One Line
Attachment 12C – Nellis AFB – Model
Attachment 12D – Nellis AFB – Study
Attachment 12E – Nellis AFB – Generator

Attachment 13A – MacDill AFB – Distribution Map
Attachment 13B – MacDill AFB – One Line
Attachment 13C – MacDill AFB – Model
Attachment 13D – MacDill – Study
Attachment 13E – MacDill – Generator

Attachment 14A – Moody AFB – Distribution Map
Attachment 14B – Moody AFB – One Line
Attachment 14C – Moody AFB – ETAP Model
Attachment 14D – Moody AFB – Study
Attachment 14E – Moody AFB – Generator

Attachment 15A – Patrick SFB – Distribution Map
Attachment 15B – Patrick SFB – One Line
Attachment 15C – Patrick SFB – Model
Attachment 15D – Patrick SFB – Study
Attachment 15E – Patrick SFB – Generator

Attachment 16A – Cape Canaveral SFS – Distribution Map
Attachment 16B – Cape Canaveral SFS – One Line
Attachment 16C – Cape Canaveral SFS – Model
Attachment 16D – Cape Canaveral SFS – Study
Attachment 16E – Cape Canaveral SFS – Generator

Attachment 17A – Cannon AFB – Distribution Map
Attachment 17B – Cannon AFB – One Line
Attachment 17C – Cannon AFB – Model
Attachment 17D – Cannon AFB – Study
Attachment 17E – Cannon AFB – Generator

Attachment 18A1, A2 – Fairchild AFB – Distribution Map(s)
Attachment 18B1, B2 – Fairchild AFB – One-Line(s)
Attachment 18C1, C2 – Fairchild AFB – Model(s)
Attachment 18D – Fairchild AFB – Study
Attachment 18E – Fairchild AFB – Generator

Attachment 19A – Holloman AFB – Distribution Map – Primary
Attachment 19B – Holloman AFB – One-Line
Attachment 19C – Holloman AFB – Model
Attachment 19D – Holloman AFB – Study
Attachment 19E – Holloman AFB – Generator

Attachment 20A – Peterson SFB – Distribution Map and Equipment
Attachment 20B – Peterson SFB – One-line
Attachment 20C – Peterson SFB – ETAP Model
Attachment 20D – Peterson SFB – Study
Attachment 20E – Peterson SFB – Generator

Attachment 21A – Suwon AB – Distribution Map
Attachment 21B – Suwon AB – One-Line
Attachment 21C – Suwon AB – Model
Attachment 21D – Suwon AB – Study
Attachment 21E – Reserve

Attachment 22A – Gimhae AB – Distribution Map
Attachment 22B – Gimhae AB – One-Line
Attachment 22C – Gimhae AB – Model
Attachment 22D – Gimhae AB – Study
Attachment 22E – Gimhae AB – Generator

Attachment 23A – Yokota AB – Distribution Map,
Attachment 23B – Yokota AB – One-Line
Attachment 23C – Yokota AB – Model
Attachment 23D – Yokota AB – Study
Attachment 23E – Yokota AB – APIMS Generator

Attachment 24A – Aviano AB- Distribution Map
Attachment 24B – Aviano AB - One-Line
Attachment 24C – Aviano AB - Model
Attachment 24D – Aviano AB - Study
Attachment 24E – Aviano AB - Generator

Attachment 25A – Moron AB – Distribution Map
Attachment 25B1, B2 – Moron AB – One-Line (s)
Attachment 25C1, C2 – Moron AB – Model(s)
Attachment 25D – Moron AB – Study
Attachment 25E – Moron AB – Generator

Attachment 26A – Misawa AB – Distribution Map

Attachment 26B – Misawa AB – One-Line
Attachment 26C1, C2 – Misawa AB – Model(s)
Attachment 26D – Misawa AB – Study
Attachment 26E – Misawa AB – Generator

Attachment 27A1, A2, A3 – Incirlik AB – Distribution Map(s)
Attachment 27B1 & B2 – Incirlik AB – One-Line(s)
Attachment 27C – Incirlik AB – Model ETAP
Attachment 27D – Reserve
Attachment 27E – Incirlik – Generator

Attachment 28A – Clear AB – Distribution Map
Attachment 28B – Clear AB – One-Line
Attachment 28C – Clear AB – Model
Attachment 28D – Clear AB – Study
Attachment 28E – Clear AB – Generator